

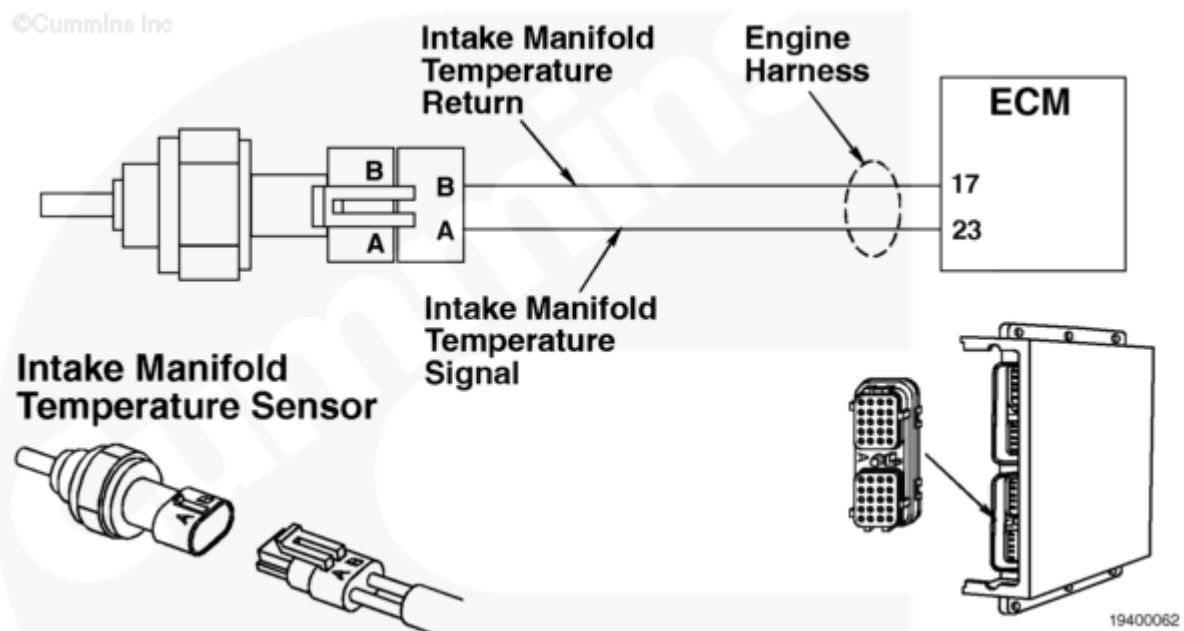
Fault Code: 155

Intake Manifold Temperature Sensor - Engine Protection

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Overview

Codes	Reason	Effect
Fault Code: 155 PID(P): P105 SPN: 105 FMI: 0 Lamp: Engine Protection SRT: 00-360	High intake air manifold temperature has been detected. Voltage signal at intake manifold air temperature signal pin 23 indicates intake manifold air temperature above 104°C [219°F].	Calibration-dependent progressive power and speed derate and engine shutdown as the temperature increases over thresholds.



Intake Manifold Temperature Sensor Circuit

Circuit Description

The intake manifold temperature sensor is used by the ECM to monitor the temperature of the air in the intake manifold after the aftercooler. The intake manifold temperature sensor is used by the ECM for the engine protection system, timing, and fueling control. The ECM monitors the

voltage on pin 23. The ECM expects to see the voltage vary between 0.5 and 4.5-VDC. If the voltage is below 0.5-VDC, the ECM will log Fault Code 155. Voltage below 0.5-VDC on pin 23 can be caused by a cooling system malfunction or an in-range sensor malfunction.

Component Location

QSK19 series engines - The intake manifold temperature sensor is located above the fuel pump, next to the intake manifold pressure sensor.

Shoptalk

The following chart shows resistance of the intake manifold temperature sensor at various temperature readings.

Temperature(°C)	Temperature[°F]	Resistance(ohms)
0	32	30k to 36k
25	77	9k to 11k
50	122	3k to 4k
75	167	1350 to 1500
100	212	600 to 675

Refer to Troubleshooting Fault Code 155. (</qs3/pubsys2/xml/en/procedures/19/19-t05-155.html>)